

SHUBAN M S

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SUMMARY

AI Engineer with experience building production-grade agentic AI systems, distributed observability infrastructure, and LLM evaluation tooling on Azure. Strong at designing multi-agent pipelines with LangGraph, instrumenting microservices end-to-end with OpenTelemetry, and turning ambiguous problems into reliable systems with real business impact.

EXPERIENCE

CGI

June 2025 – Present

AI Engineer & SDE-1

Bengaluru, India

- Designed a **multi-stage LangGraph agentic pipeline** (5 agents: Manager, OCR, Reviewer, Enhancer, Review-2) for medical document processing on Azure, achieving **~97% extraction accuracy** — 93% improvement over traditional OCR — at **~\$1/document**, cutting costs by **~80%**.
- Engineered a **cross-service distributed observability framework** (OpenTelemetry, Jaeger, Prometheus, Grafana) with end-to-end trace propagation across Azure Service Bus and ProcessPoolExecutor boundaries; custom dashboards track LLM token usage per agent, OCR throughput, and queue depth.
- Architected **RECON** (Real-world Evaluation, Compliance & Observability Network) — an internal AI eval platform covering RAG, Agent, and LLM modules with an **EU AI Act / ISO 42001 compliance layer**; submitted to CGI FY2026 Innovation Survey.
- Built a **black-box RAG validation pipeline** for medical APS documents with 5 QGen agents (factual, inferential, adversarial, negation, multi-hop) and **BioClinical ModernBERT** embeddings.

GreenStitch.io

Aug 2023 – Jan 2024

Founding ML & Backend Engineer

Bengaluru, India

- Owned and built a **FastAPI-based Python backend** serving real-time and batch inference for **Scope 3 carbon emissions prediction**.
- Designed normalized relational schemas and built a **vector-based semantic search engine** using NLP techniques, achieving **99% retrieval accuracy**.
- Developed stacked ensemble models using TensorFlow and Keras with clean ETL pipelines, consistently achieving **99%+ accuracy and strong F1 scores** in production.

Vellore Institute of Technology

June 2024 – Aug 2024

Lead Data Scientist – Research

Chennai, India

- Collected a medical gait dataset using an Xsens MTw Awinda IMU kit, capturing stair ascent and descent motion data.
- Built Python pipelines for IMU signal cleaning and feature extraction across time and frequency domains.
- Applied unsupervised clustering techniques, with agglomerative clustering performing best for weight-based classification.

PATENTS & PUBLICATIONS

2 Patents Filed – IN 202541065615: Color-Dot-Based Authentication (Jul 2025) & IN 202541016033: GenAI Learning System with Blockchain Traceability (Feb 2025), VIT Chennai.

IEEE Best Paper Award – FALCONNet: Multi-Defense Approach for Securing Few-Shot Learning Against Adversarial Attacks; Springer publication on YOLOv8 Parking Lot Occupancy with PSAT Algorithm.

EDUCATION

Vellore Institute of Technology

2021 – 2025

B.Tech in Electronics and Computer Engineering (CGPA: 8.48)

Chennai, India

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

Agentic AI: LangGraph, LangChain, RAG Pipelines, LLM Evaluation, OpenInference, Agentic AI

Observability: OpenTelemetry, Jaeger, Prometheus, Grafana, Distributed Tracing

Backend & Cloud: FastAPI, Azure (Blob Storage, Service Bus, Azure OpenAI), Docker, Microservices

Data & ML: PostgreSQL, PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, NumPy

Tools & Compliance: Git, GitHub Actions, Docker, EU AI Act, ISO/IEC 42001